



LECTURE 7

Sprint Execution, Review and Retrospective

COEN / ELEC 390

Acknowledgement

- The concepts in these slides are based on the textbook:

Essential Scrum, A Practical Guide to the Most Popular Agile Process
Kenneth S. Rubin, Addison-Wesley, 2013.

Sprint Execution

- During sprint execution the development team members self-organize and determine the best way to meet the goal established during sprint planning.
- The ScrumMaster participates as the coach, facilitator, and impediment remover.
- The product owner must be available during sprint execution to answer clarifying questions, to review intermediate work and provide feedback to the team.
- On a two-week-long sprint, sprint execution might account for about eight out of the 10 working days.

Flow Management

- It is **the team's responsibility to manage the flow of work** during sprint execution to meet the sprint goal.
- It must **make decisions** such as how much work the team should do in parallel, when work should begin on a specific item, how the task-level work should be organized, etc.
- We should **discard old habits** such as:
 - Feeling the need to keep everyone constantly occupied.
 - Assuming that tasks must be completed in a strict sequence.
 - Encouraging individuals to focus solely on their specific roles in the solution.

Guidelines

- Just as working on too many tasks at once can be counterproductive, so can focusing on too few tasks at a time
- Support one another by collaborating on tasks that need completion before starting new items.
- A sprint execution is not a mini waterfall where activities are done sequentially.
- We should never call untested features done; done should mean DONE.

How to organize the tasks?

- The organization of the work **should be left to the developers**. The same approach will not always be right for every backlog item, nor for every individual.
- One can even do **test-first development**, where tests are written before the development is performed, the “testers” are the first to work on a feature
- **Ultimately the team decides**. Product owners and managers must trust that the team members are responsible professionals.

Who does the work?

- An obvious answer is **the person best able to quickly and correctly get it done.**
- What if that person is unavailable? It is **the collective responsibility** of the team members to decide.
- When team members have **T-shaped skills**, several people on the team have the ability to work on each task.
- The daily scrum is a critical, **daily inspect-and-adapt activity** to help the team achieve faster, more flexible flow toward the solution.
- It is good practice to have **a dashboard of the tasks** that need to be done.

Sprint Burndown Chart

- The sprint burndown chart is used to estimate the effort remaining each day.

Tasks	D1	D2	D3	D4	D5	D6	D7	D8	D9	...	D15
Task 1	8	4	4	2							
Task 2	12	8	16	14	9	6	2				
Task 3	5	5	3	3	1						
Task 4	7	7	7	5	10	6	3	1			
Task 5	3	3	3	3	3	3	3				
Task 6	14	14	14	14	14	14	14	8	4		
Task 7						8	6	4	2		
Tasks 8–30	151	139	143	134	118	99	89	101	84		0
Total	200	180	190	175	155	130	115	113	90		0

Source: Essential Scrum, A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin, Addison-Wesley, 2013.

Sprint Review

- During the sprint review, we **evaluate and adapt** the results of the work, focusing on the potentially shippable product increment.
- The review offers **a clear view** of the product's current status, **revealing any challenges**.
- It is an **opportunity to ask questions, share observations, suggest improvements**, and engage in discussions to ensure we're heading in the right direction.
- The sprint review is a **crucial learning loop** within the Scrum framework.

Participants

- The sprint review provides an important opportunity for the Scrum team to get feedback from people who typically are not available on a daily basis during sprint execution.
- Who should attend?
 - Scrum team, internal stakeholders (business owners, executives, managers, etc.), other internal teams (marketing, sales, legal, etc.), external stakeholders (customers, users, etc.)

Sprint Review Process

- A Scrum team member, often the product owner, **summarizes** the sprint goal, associated PBIs, and the achieved product increment, assessing whether the results meet the goal.
- The sprint review prioritizes conversations and collaboration, allowing Scrum team members to **demonstrate** the product increment and shows tests that confirm the work meets the PO's satisfaction.
- The sprint review identifies issues and potential solutions through **discussion**, but deep problem-solving should occur in a separate meeting, allowing non-Scrum members to ask questions and guide the product's direction.
- The team addresses stakeholder feedback, evaluates the relevance of the product, identifies missing features, and assesses resource allocation, often leading to grooming of the backlog and potential adjustments to release scope, date, or budget (**adaption**).

Sprint Retrospective

- The sprint retrospective is one of the most important and least appreciated practices in the Scrum framework.
- It is important because it gives teams the chance to customize Scrum to their unique circumstances.
- It is underappreciated because some people have a misguided view that it takes time away from doing “real” work.
- Because retrospectives occur at the end of each sprint, it can apply early and incremental learning, significantly affecting the outcome of the project.

Sprint Retrospective (cont.)

- A sprint retrospective can be as simple as the Scrum team members coming together to discuss questions such as:
 1. What worked well this sprint that we want to continue doing?
 2. What didn't work well this sprint that we should stop doing?
 3. What should we start doing or improve?
- Based on their discussions, team members determine a few actionable changes to improve the 'process' of the next sprint.

Participants

- The full Scrum team attends:
 - The ScrumMaster must attend because of their integral role as process experts
 - Some argue that having the product owner at the retrospective might inhibit the team from being completely honest or revealing difficult issues.
- Stakeholders or managers who are not on a Scrum team, on the other hand, should attend a retrospective only if invited by the Scrum team

Sprint Retrospective Steps

1. **Define the retrospective focus:** The default focus is to review all relevant aspects of the process, but a team may decide to have a different focus (e.g., provide training, etc.)
2. **Select the exercises:** A typical retrospective includes creating and mining a sprint event timeline, brainstorming insights, grouping and voting on insights.
3. **Gather objective data (not opinions):** What events happened and when, or counts of the number of PBIs that were started but not finished, or the feature burnup chart for the sprint illustrating the flow of completed work.
4. **Structure the Retrospective:** Sprint retrospectives should consistently occur at the same location, day, and time each sprint, with their length varying based on team size, experience, and remote participation.

Sprint Retrospective Steps (Cont.)

5. **Set the atmosphere:** A safe environment with clear ground rules encourages open dialogue, while personnel issues should be managed separately; fostering active participation is key for effective retrospectives
6. **Share context:** A group of people can all experience the same event and yet interpret it quite differently. To successfully inspect the current sprint, it is important to get everyone on the same page so that they have a shared context.
7. **Identify insights:** the participants thoughtfully examine, understand, and interpret the data to identify process improvement insights: What worked well? What didn't work well? Where are some opportunities to do things differently?
8. **Determine actions:** Prioritize insights through majority voting, determine action steps, and create an improvement backlog for issues identified in retrospectives that can't be addressed right away.
9. **Follow through:** To ensure that what happens in the sprint retrospective does not just stay in the sprint retrospective, the participants should follow up on the actions they chose to complete.